Definition of the differential: If $\frac{dy}{dx} = f'(x)$ then $dy = dx \cdot f'(x)$

1) Find the indefinite integral $\int \cos(4x) \cdot dx$

Let $u = 4x$

a) $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) By definition of Differential: $du = \underline{\quad ? \quad}$

c) $\dfrac{1}{4} \cdot du = \underline{\quad ? \quad}$

d) Write the integral $\int \cos(4x) \cdot dx$ in terms of u and then evaluate.

$$\int \cos(4x) \cdot dx = \int \underline{\quad ? \quad} \cdot du = \underline{\quad ? \quad}$$

e) $\int \cos(4x) \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)

2) Find the indefinite integral $\int \sin x \cdot \cos x \cdot dx$

Let $u = \sin x$

a) $\dfrac{du}{dx} = $ __?__

b) By definition of Differential: $du = $ __?__

c) Write the integral $\int \sin x \cdot \cos x \cdot dx$ in terms of u and then evaluate.

$\int \sin x \cdot \cos x \cdot dx = \int$ __?__ $\cdot du = $ __?__

d) $\int \sin x \cdot \cos x \cdot dx = $ __?__ (Note: Answer must be in terms of x.)

3) Find the indefinite integral $\int \left(1-\cos^2 x\right)\cdot \sin x \cdot dx$

Let $u = \cos x$

a) $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) By definition of Differential: $du = \underline{\quad ? \quad}$

c) $-1 \cdot du = \underline{\quad ? \quad}$

d) Write the integral $\int \left(1-\cos^2 x\right)\cdot \sin x \cdot dx$ in terms of u and then evaluate.

$$\int \left(1-\cos^2 x\right)\cdot \sin x \cdot dx = \int \underline{\quad ? \quad} \cdot du = \underline{\quad ? \quad}$$

e) $\int \left(1-\cos^2 x\right)\cdot \sin x \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)

4) Find the indefinite integral $\int \tan x \cdot \sec^2 x \cdot dx$

Let $u = \tan x$

a) $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) By definition of Differential: $du = \underline{\quad ? \quad}$

c) Write the integral $\int \tan x \cdot \sec^2 x \cdot dx$ in terms of u and then evaluate.

$$\int \tan x \cdot \sec^2 x \cdot dx = \int \underline{\quad ? \quad} \cdot du = \underline{\quad ? \quad}$$

d) $\int \tan x \cdot \sec^2 x \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)

5) Find the indefinite integral $\int \csc^2 x \cdot \cot x \cdot dx$

Hint: $\int \csc^2 x \cdot \cot x \cdot dx = \int \csc x \cdot \csc x \cdot \cot x \cdot dx$

Let $u = \csc x$

a) $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) By definition of Differential: $du = \underline{\quad ? \quad}$

c) Write the integral $\int \csc x \cdot \csc x \cdot \cot x \cdot dx$ in terms of u and then evaluate.

$$\int \csc x \cdot \csc x \cdot \cot x \cdot dx = \int \underline{\quad ? \quad} \cdot du = \underline{\quad ? \quad}$$

d) $\int \csc x \cdot \csc x \cdot \cot x \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)

6) Find the indefinite integral $\int \cos(\pi x) \cdot dx$

Let $u = \pi \cdot x$

a) $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) By definition of Differential: $du = \underline{\quad ? \quad}$

c) $\dfrac{1}{\pi} \cdot du == \underline{\quad ? \quad}$

d) Write the integral $\int \cos(\pi x) \cdot dx$ in terms of u and then evaluate.

$$\int \cos(\pi x) \cdot dx = \int \underline{\quad ? \quad} \cdot du = \underline{\quad ? \quad}$$

e) $\int \cos(\pi x) \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)